About this PDF

Built only from: **NOC:Computer Networks and Internet Protocol (NPTEL course book, weeks 8, 9)** — https://nptel.ac.in/courses/106105183

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Covers 6 of 8 syllabus sub-topics of this unit.

Partly covered in the source:

- Routing and protocols: Unicast routing

Not in the source (not covered in this PDF):

- Multicast Routing: DVMRP
- PIM.

Routing and protocols: Unicast routing

Partly covered in the source.

Definition

Unicast routing is the process by which a router determines a single optimal path for each packet to reach its destination and forwards the packet along that path, using routing protocols that build and maintain a routing table.

Explanation

Dynamic routing protocols automatically discover and maintain awareness of network paths, allowing routers to update their routing tables so that packets are forwarded optimally. Intra-domain routing uses two main categories of protocols: distance-vector (e.g., RIP) and link-state (e.g., OSPF). These protocols periodically execute, construct the routing table, and provide the destination-to-next-hop mapping needed for forwarding. When a packet arrives, the router extracts the destination IP address from the IP header, looks up this address in the routing table (or forwarding information base), determines the appropriate next-hop interface, and forwards the packet accordingly. The routing information base (RIB) holds the complete set of routes in software at the control plane, while the forwarding information base (FIB) is a hardware copy (e.g., in TCAM) used for fast look-ups during packet forwarding. The RIB feeds the FIB, and updates are copied as needed.

Example

A router’s routing table may contain entries such as: destination A (cost 0), B (cost 5), C (cost 2), D (cost 3), and E (cost 6 via C). This table is constructed from information exchanged by a link-state protocol like OSPF and is then used to forward packets to the appropriate next hop.

Algorithm / Steps

1. Execute a routing protocol (distance-vector or link-state) to discover paths and construct the routing table.
2. Store the complete routing information in the RIB at the control plane.

3. Copy required routes from the RIB to the FIB (hardware) for fast lookup.
4. Upon receiving a packet, extract the destination IP address from the IP header.
5. Perform a route lookup in the FIB to find the matching entry and determine the next-hop interface.
6. Forward the packet to the identified next-hop interface.

Key points

- Dynamic routing protocols (distance-vector and link-state) automatically build and update routing tables for optimal unicast forwarding.
- The router's control plane maintains the RIB, while the forwarding plane uses the FIB for fast packet forwarding.
- Forwarding involves extracting the destination IP, looking up the next hop in the routing/FIB table, and sending the packet out the appropriate interface.
- Intra-domain protocols such as RIP (distance-vector) and OSPF (link-state) are commonly used for unicast routing within an autonomous system.

Distance Vector Routing

Definition

Distance vector routing is a type of intra-domain dynamic routing protocol in which each router maintains a distance (or cost) to each destination and shares this information only with its immediate neighbors. The path with the smallest cost is preferred for reaching a destination, and the information is kept in a distance-vector table that is periodically advertised to neighbors.

Explanation

In distance vector protocols, a router does not have a complete view of the entire network. Instead, it relies on the cost information received from its directly connected neighbors. Each router updates its own distance-vector table based on the advertisements from neighbors and selects the minimum-cost path to each destination. This localized exchange of cost information allows routers to dynamically adjust to changes such as added or removed links, keeping routing tables up-to-date.

Example

- Destination A: cost 0
- Destination B: cost 5
- Destination C: cost 2
- Destination D: cost 3
- Destination E: cost 6 via C

Algorithm / Steps

1. Each router maintains a distance-vector table containing the cost to each known destination.

2. The router periodically advertises its distance-vector table to all directly connected neighbors.
3. Upon receiving a neighbor's advertisement, the router adds the cost of the link to that neighbor to the received costs and updates its own table if a lower-cost path is found.
4. The router selects the path with the smallest total cost for each destination and uses this information for forwarding packets.

Key points

- Routers share only distance (cost) information with immediate neighbors.
- The smallest-cost path is chosen as the preferred route.
- Distance-vector tables are updated and advertised periodically.
- This approach provides a localized, dynamic method for maintaining routing tables within an autonomous system.

RIP

Definition

RIP (Routing Information Protocol) is a popular distance-vector routing protocol used for intra-domain routing within an autonomous system.

Explanation

In distance-vector routing each router maintains a routing table that lists the cost to reach each destination. Routers periodically share their routing tables with immediate neighbors. When a router receives an update, it recomputes the best cost to each destination based on the information received. This process continues until all routers have consistent tables, allowing packets to be forwarded along the lowest-cost paths.

Example

Consider router A that initially knows only its directly connected networks: cost to itself is 0, cost to B is 5, cost to C is 2, cost to D is 3, and it does not know a path to E (treated as infinity). After receiving an update from neighbor C, which knows a path to E with cost 4, router A updates its table to reach E via C with a total cost of 6 (2 + 4).

Algorithm / Steps

1. Each router initializes its routing table with costs to directly connected neighbors and sets unknown destinations to a large value (infinity).
2. At regular intervals, a router sends its entire routing table to all immediate neighbors.
3. Upon receiving a neighbor's table, the router adds the cost to reach that neighbor to each advertised cost and compares the result with its current entry.
4. If the new cost is lower, the router updates its table with the new cost and the next-hop neighbor.

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5. The updated table is then used for forwarding packets and is advertised in the next periodic update.

Key points

- RIP is a distance-vector protocol that relies on periodic exchange of complete routing tables with neighbors.
- It requires only local state and has a small memory footprint, but can suffer from routing loops.
- Updates are triggered periodically or when a change is detected, allowing routers to converge on the shortest-cost paths.
- The protocol is commonly used as an interior gateway protocol (IGP) within an autonomous system.

Link State Routing

Definition

Link state routing is an intra-domain routing approach in which each router maintains a description of its interfaces—including IP address, subnet mask, network type and connectivity to neighboring routers—and shares this information to build a network-wide topology view.

Explanation

Each router creates a **link-state database** that contains the collection of link-state advertisements (LSAs) received from all routers in the network. Using this database, routing algorithms compute the most desirable routes to every destination, and the resulting routes are installed in the routing table. Thus every router has a consistent view of the whole network topology and can determine optimal paths based on that view.

Example

In OSPF (Open Shortest Path First), routers exchange LSAs of various types—router-link, network-link, summary-link to network, summary-link to AS-boundary router, and external-link—to disseminate link-state information throughout the autonomous system.

Algorithm / Steps

1. **Identify directly-connected links** – each router determines the routers and networks on its immediate interfaces.
2. **Generate LSAs** – the router creates link-state advertisements describing these links and their associated costs.
3. **Distribute LSAs** – LSAs are flooded to all other routers in the domain.
4. **Build link-state database** – every router collects LSAs from all neighbors and assembles a complete topology database.
5. **Compute shortest paths** – using the topology database, the router runs a shortest-path algorithm to find the optimal route to each destination.

6. **Update routing table** – the computed routes are entered into the routing table for packet forwarding.

Key points

- Link state routing maintains a full network graph by sharing interface state information among routers.
- LSAs are the mechanism by which routers advertise their directly-connected links and costs.
- All routers end up with an identical link-state database, enabling consistent route computation.
- OSPF is a widely used link-state protocol that defines multiple LSA types for intra-AS information dissemination.

OSPF Path-vector routing

Definition

Path-vector routing is a routing method used for inter-domain routing where the path is represented as an ordered list of autonomous systems (ASes) that a packet must traverse. Each AS router advertises reachability of its networks to neighboring AS boundary routers using path-vector messages, and routers verify these advertised paths against defined policies before accepting them. In the broader classification of routing protocols, intra-domain routing employs distance-vector or link-state protocols (e.g., OSPF), while inter-domain routing relies on path-vector protocols such as BGP.

Explanation

A path-vector message carries the sequence of AS identifiers that the route has passed through. When a router receives such a message, it checks whether the advertised path complies with its routing policies (for example, disallowing routes that contain a particular AS) before installing the route in its table. This policy-based verification enables routers to enforce routing decisions based on administrative preferences, not just on metric calculations. The periodic exchange of these messages among neighboring AS boundary routers builds a consistent view of reachable networks across autonomous systems.

Key points

- Path-vector routing represents a route as an ordered list of autonomous systems that the packet must traverse.
- Each AS boundary router advertises network reachability to its neighbors using path-vector messages.
- Received paths are validated against local routing policies, allowing policy-based routing decisions.
- Inter-domain routing uses path-vector protocols (e.g., BGP), while intra-domain routing uses distance-vector or link-state protocols such as OSPF.

BGP

Definition

Border Gateway Protocol (BGP) is an exterior routing protocol that employs the path-vector routing approach for inter-domain (inter-AS) routing. It is used to exchange routing information between autonomous systems (AS) and is based on a set of well-known attributes that describe the advertised paths.

Explanation

BGP operates by maintaining a database of reachable networks and the preferred routes to those networks. It uses four types of messages—**OPEN**, **UPDATE**, **KEEPALIVE**, and **NOTIFICATION**—which are exchanged over a TCP connection on port 179. The OPEN message initiates communication with a peer, UPDATE messages create, modify, or withdraw route advertisements, KEEPALIVE messages confirm that a router is still reachable, and NOTIFICATION messages signal errors.

In the path-vector model, each route advertisement carries the **ordered list of autonomous systems** that a packet must traverse to reach the destination. This differs from distance-vector routing, which only conveys the next hop and a metric; the AS-path information enables BGP to enforce routing policies such as avoiding certain networks or preferring others,.

Well-known attributes (some mandatory, some discretionary) are attached to each route advertisement. These attributes allow routers to fine-tune path selection and to propagate necessary updates to peers.

Example

Consider a router **N1** that originates a route for network **N1**. The next hop for this route is router **R1**, which belongs to a particular autonomous system. The BGP update from N1 to R1 would include the AS-path indicating that traffic to network N1 must pass through R1’s AS.

Key points

- BGP uses four message types (OPEN, UPDATE, KEEPALIVE, NOTIFICATION) over TCP port 179.
- It is a path-vector protocol that carries the full AS-path for each advertised route.
- Routing decisions can be based on policies (e.g., avoid or prefer certain ASes) in addition to simple metrics.
- Well-known attributes, some mandatory and some discretionary, define and update path characteristics.

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